

EXPLAINING MONOCULAR DEPTH ESTIMATION: A MULTI-METHOD XAI ANALYSIS OF DEPTH ANYTHING V2

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ABSTRACT

Monocular depth networks produce a dense geometric prediction from a single RGB image, but a low aggregate error does not reveal which visual evidence drives that prediction. We study the pretrained Depth Anything V2 Large model on NYU Depth V2 using five complementary explanation views: Attention Rollout, a TCAV-style concept activation vector (CAV) projection, Integrated Gradients, decoder Grad-CAM++, and Occlusion Sensitivity. Predictions are evaluated on 500 test images after per-image scale-shift alignment; detailed explanations are generated for 12 good, random, and failure cases. Besides qualitative overlays, we measure alignment with depth boundaries and error regions, deletion faithfulness against random masking, parameter-randomization sensitivity, runtime, and GPU memory. The model obtains mean RMSE 0.556 and $\delta_1 = 0.810$. Integrated Gradients best localizes high-error regions, while CAV projection yields the largest deletion gap but also high variance. All methods exhibit weak ground-truth edge correlation, showing that visually plausible saliency is not sufficient evidence of geometric reasoning.

1 INTRODUCTION

Monocular depth estimation maps one RGB image to a depth value at every pixel. It is useful for robotics, navigation, augmented reality, and three-dimensional scene understanding, but it is intrinsically ambiguous: different three-dimensional scenes can produce similar two-dimensional projections. Modern foundation models such as Depth Anything V2 (DAV2) reduce prediction error by learning broad visual priors from large image collections (Yang et al., 2024). Their accuracy, however, does not make their reasoning transparent. A depth map may look smooth and plausible while relying on texture, illumination, or dataset-specific cues rather than stable scene geometry.

This issue is especially important for dense regression. Most attribution methods were introduced for a class logit, whereas a depth network outputs hundreds of thousands of continuous values. An explanation therefore depends on both the attribution method and the scalar objective extracted from the depth map. Moreover, a colorful heatmap can appear meaningful even when it is insensitive to model parameters (Adebayo et al., 2018). We consequently treat explanation as an evaluation problem rather than relying on visual inspection alone.

We ask four questions: (1) do explanations align with ground-truth depth boundaries rather than only RGB texture; (2) do they identify regions where the model makes large errors; (3) does masking highly attributed input content change the prediction more than random masking; and (4) do explanations change when learned parameters are randomized? We answer these questions with five XAI approaches spanning transformer attention, input gradients, decoder activations, semantic concepts, and direct perturbations. Our main contributions are: a reproducible dense-depth XAI pipeline; quantitative geometry, error, faithfulness, and sanity metrics; and a comparison of explanatory value against runtime and GPU-memory cost.

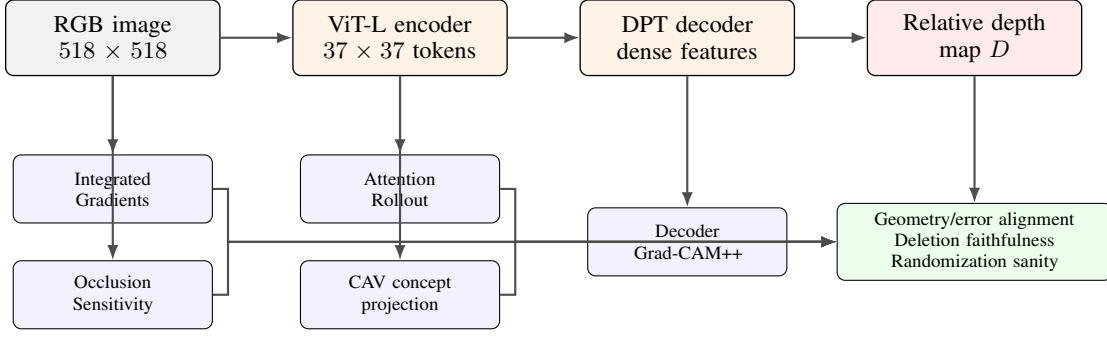


Figure 1: Evaluation pipeline. Explanations are extracted at input, encoder, or decoder level and assessed against ground-truth geometry, model errors, perturbation response, and parameter dependence.

2 METHODS

2.1 DEPTH PREDICTION MODEL

We use the pretrained DAV2 Large model with a ViT-L encoder and DPT-style spatial decoder (Yang et al., 2024; Ranftl et al., 2021). The encoder is initialized from DINOv2 representations (Oquab et al., 2024); patch size 14 converts a 518×518 input into a 37×37 patch grid. The complete model has 335.3M parameters. It predicts relative depth, so raw values are not directly comparable with metric NYU depth.

For each image, predicted depth d_i is aligned to ground truth g_i using least-squares scale and shift:

$$(s^*, t^*) = \arg \min_{s, t} \sum_{i \in \mathcal{V}} (s d_i + t - g_i)^2, \quad \hat{d}_i = s^* d_i + t^*, \quad (1)$$

where $\mathcal{V}$ contains valid pixels. This evaluates relative-depth structure; it must not be interpreted as zero-shot metric-depth performance.

Dense attribution requires a scalar target. We explain the global mean of the unaligned network output,

$$T(D) = \frac{1}{HW} \sum_{i=1}^H \sum_{j=1}^W D_{ij}, \quad (2)$$

so every method answers which input regions most influence the image-level mean depth response. The alignment in Eq. 1 is used only for prediction metrics and error maps, not inside the attribution objective.

2.2 EXPLANATION METHODS

Attention Rollout. For each ViT block, attention is averaged over heads, augmented with the identity residual, and row-normalized. The matrices are multiplied across blocks to propagate token influence (Abnar & Zuidema, 2020). The patch-to-patch relevance is averaged and bilinearly upsampled to image resolution. This is a fast global view of transformer information routing, but attention is not itself a causal attribution.

Integrated Gradients (IG). IG integrates the gradient of T along a straight path from a blurred baseline x' to the image x (Sundararajan et al., 2017):

$$\text{IG}(x) = (x - x') \odot \frac{1}{m} \sum_{k=1}^m \nabla_x T \left(x' + \frac{k}{m} (x - x') \right). \quad (3)$$

We use $m = 16$ steps, sum absolute attributions across RGB channels, and normalize the result. IG offers pixel-level sensitivity but depends on the baseline and incurs repeated forward/backward passes.

Decoder Grad-CAM++. Grad-CAM++ weights spatial feature maps using higher-order gradient information (Chattopadhyay et al., 2018). We automatically select the deepest usable spatial decoder layer and explain the same target $T(D)$. It is cheaper than IG and directly probes the features that construct the depth map, although its resolution and interpretation depend on the selected layer.

Occlusion Sensitivity. We replace 74×74 image regions, at stride 74, with the same blurred baseline used by IG. The score assigned to a patch is $|T(D(x)) - T(D(x_{occ}))|$. Overlapping contributions are averaged. Unlike gradient methods, occlusion directly tests input intervention, but its maps are coarse and computation grows with the number of patches.

TCAV-style CAV projection. Following the concept-vector idea of TCAV (Kim et al., 2018), we define three depth-derived concepts: sharp depth edges, flat regions, and slope changes. From 48 training candidates, the six highest- and six lowest-scoring examples per concept form positive and negative sets. A linear classifier on normalized final-block ViT patch activations yields a CAV. The CAV is reshaped to the patch grid and projected patch-wise onto test activations; the three normalized maps are averaged. This is a spatial CAV projection, *not* standard TCAV: we do not compute output directional derivatives, repeated random-concept baselines, or significance tests.

2.3 XAI EVALUATION

We compare each heatmap with Canny RGB edges, Sobel ground-truth depth edges, Sobel predicted-depth edges, and absolute aligned error. Spearman correlation measures global monotonic agreement. Top-10 overlap is $|A_{10} \cap B_{10}|/|A_{10}|$, where each set contains the highest-scoring 10% of pixels.

Faithfulness is measured by masking the top-attributed pixels at fractions $\{.02, .05, .10, .20, .30\}$ with a blurred baseline. We report the mean target change under top masking minus the corresponding random-mask mean (AOPC gap); a positive value indicates that attributed regions matter more than random regions. Finally, a cascading sanity check reinitializes the decoder, the last four encoder blocks, or the full network and recomputes IG and Grad-CAM++ (Adebayo et al., 2018). Explanation Spearman correlation and top-10 overlap quantify dependence on learned parameters.

3 EXPERIMENTAL SETTINGS

Dataset and sampling. NYU Depth V2 provides 1,449 aligned indoor RGB-depth pairs (Silberman et al., 2012). The full dataset is divided into two non-overlapping subsets, each with its own concept/reference and test split. Subset 1 uses indices 0–794 as the concept/reference split and indices 795–1448 as test data. Subset 2 uses indices 1449–2243 as the concept/reference split and indices 2244–2897 as test data. Our pipeline is designed to operate on both subsets and produces results for each independently; however, this report presents and discusses the results of Subset 1 only. Prediction metrics are computed on the first 500 test images. For detailed XAI, the three lowest-RMSE and three highest-RMSE cases are combined with six fixed-seed random cases, yielding 12 images. The extreme groups support failure analysis but are not representative estimates; the random group supplies less biased qualitative coverage.

Environment. Experiments ran with Python 3.12.13, PyTorch 2.11.0+cu128, CUDA 12.8, and an NVIDIA A100-SXM4 40GB. Input size was 518^2 , seed 42, and valid depth was clipped to $[10^{-3}, 10]$ m. The complete pipeline took 838.2 s, including 11.5 s for concept preparation.

Prediction metrics. We report absolute relative error (AbsRel), root mean squared error (RMSE), mean absolute error (MAE), and threshold accuracies $\delta_k = \Pr(\max(\hat{d}/g, g/\hat{d}) < 1.25^k)$ for $k \in \{1, 2, 3\}$, standard measures for monocular depth estimation (Eigen et al., 2014).

Table 1: Aligned depth performance. Overall values are mean $\pm$ standard deviation over 500 test images. Good/bad rows are deliberately selected extremes; random contains six fixed-seed samples.

Subset	AbsRel $\downarrow$	RMSE $\downarrow$	MAE $\downarrow$	$\delta_1 \uparrow$	$\delta_2 \uparrow$	$\delta_3 \uparrow$
Overall (500)	.159 $\pm$.074	.556 $\pm$.265	.408 $\pm$.212	.810 $\pm$.171	.951 $\pm$.057	.972 $\pm$.026
Good (3)	.044 $\pm$.008	.115 $\pm$.028	.074 $\pm$.022	.983 $\pm$.007	.996 $\pm$.004	1.000 $\pm$.000
Random (6)	.226 $\pm$.125	.709 $\pm$.332	.532 $\pm$.283	.674 $\pm$.266	.908 $\pm$.102	.943 $\pm$.054
Bad (3)	.355 $\pm$.080	1.482 $\pm$.029	1.206 $\pm$.036	.383 $\pm$.095	.809 $\pm$.128	.894 $\pm$.074

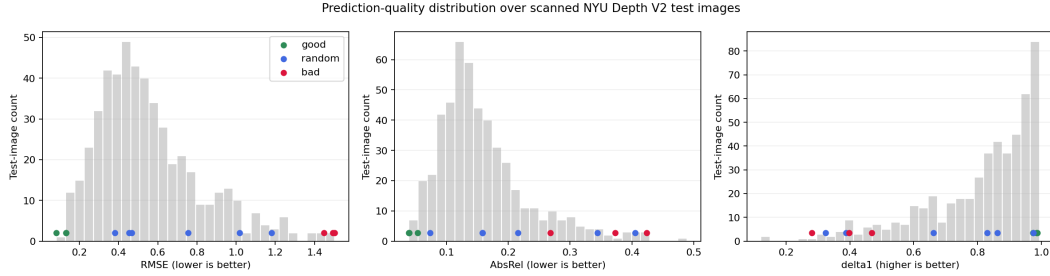


Figure 2: Prediction metric distributions for the 500-image scan. Colored markers show the good, random, and bad examples selected for explanation.

4 EXPERIMENTAL RESULTS

4.1 DEPTH PREDICTION QUALITY

Table 1 shows strong performance on many indoor images but a broad error distribution. The median RMSE is 0.502, below the mean 0.556, indicating a high-error tail. Figure 2 shows where selected examples lie in this distribution. Because alignment is fitted independently per image, failures mainly reflect incorrect spatial structure rather than a single global scale error.

Figure 3 clarifies the error modes. Fine object silhouettes can remain plausible even when large planar surfaces are misplaced. In sample 1240, the dominant error covers the back wall and windows, while the sofa remains structurally recognizable. Thus, an explanation should identify both localized object evidence and broad contextual cues.

4.2 QUANTITATIVE XAI COMPARISON

No method has strong average correlation with ground-truth depth edges (Table 2). CAV projection is highest at only 0.037; the negative means for Attention Rollout, Grad-CAM++, and occlusion caution against equating bright overlays with geometric boundary reasoning. Top-10 error overlap gives a different ranking: IG reaches 0.236, followed by Grad-CAM++ at 0.220. These methods are therefore more useful for localizing failure-sensitive regions than for recovering exact ground-truth boundaries.

CAV projection has the largest deletion gap, 25.70, but its standard deviation is 21.44. This indicates strong intervention sensitivity on some images rather than uniformly reliable behavior. Attention Rollout is second at 9.45 and is nearly as fast as CAV projection. Occlusion is directly intervention-based yet uses 24.78 GB because all patch variants in a row are batched; its coarse grid also limits boundary localization. Figure 4 visualizes these trade-offs.

4.3 QUALITATIVE EXPLANATION BEHAVIOR

The methods expose qualitatively different scales of evidence (Figure 5). Attention Rollout concentrates on a few objects and high-contrast context, while CAV projection spreads over wall, bed, and frame structures. IG emphasizes image borders and local contours, explaining its stronger error-region overlap but also revealing baseline/border artifacts. Decoder Grad-CAM++ is smoother and highlights the bed silhouette and room perimeter. Occlusion produces easily interpretable interventions but

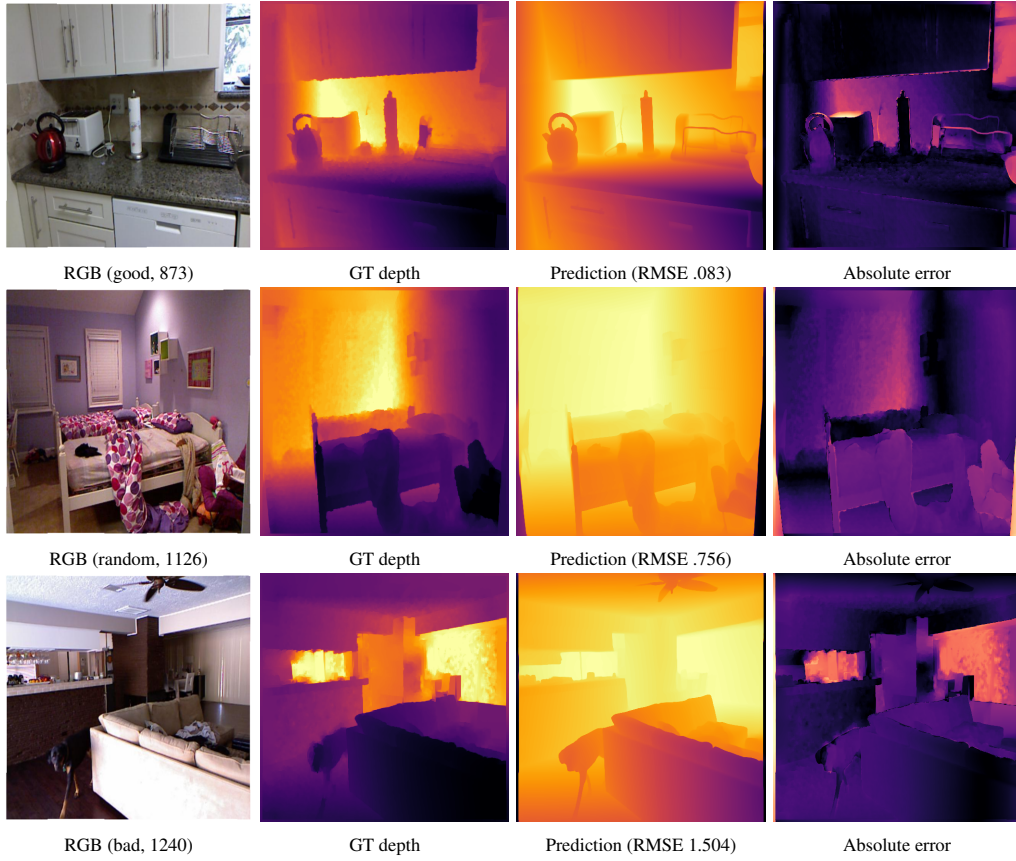


Figure 3: Qualitative depth results across the selected quality range. The good kitchen preserves object and counter ordering. The random bedroom captures coarse layout but smooths wall/bed transitions. The bad living room severely misestimates large wall and background regions despite retaining a plausible foreground sofa shape. Each row is independently color-normalized.

Table 2: XAI results over 12 images (means; standard deviation shown for AOPC). Edge correlation and overlaps are dimensionless. Runtime is per image; memory is peak allocated CUDA memory. Bold marks the best value in each explanatory metric, not a universally best method.

Method	GT-edge corr.↑	Error top-10↑	AOPC gap↑	Time (s)↓	Peak GB↓
Attention Rollout	−.009	.173	9.45 ± 15.97	.114	5.60
CAV projection	.037	.169	25.70 ± 21.44	.126	5.60
Integrated Gradients	.006	.236	4.15 ± 7.67	3.377	8.21
Grad-CAM++ decoder	−.020	.220	3.94 ± 9.04	.308	8.34
Occlusion Sensitivity	−.082	.148	6.82 ± 9.46	3.989	24.78

cannot resolve thin edges below its patch size. Agreement between methods is therefore evidence of a robust region; disagreement identifies sensitivity to explanatory assumptions.

4.4 FAITHFULNESS, SANITY, AND COMPUTATIONAL COST

The deletion curves in Figure 6 generally rise with the removed fraction. CAV projection and Attention Rollout create the largest target rise changes, suggesting that their broad regions capture features controlling the global mean depth target. In contrast, IG’s local, edge-heavy maps have higher error overlap but a smaller global deletion gap. This is not contradictory: failure localization and global-target faithfulness measure different properties.



Figure 4: Method-level geometry alignment, high-error overlap, and deletion faithfulness. The three criteria disagree, so no single heatmap should be treated as a complete explanation.



Figure 5: Five explanations for random sample 1126. Attention Rollout is sparse and object/context oriented; CAV projection is distributed; IG follows high-frequency boundaries; Grad-CAM++ forms smooth decoder-level regions; occlusion is block-like by construction.

Randomizing the decoder or the full model collapses prediction correlation to zero and explanation Spearman correlation to zero for both IG and Grad-CAM++, supporting parameter dependence. Randomizing only the last four encoder blocks leaves prediction rank correlation at 0.925; explanation correlations remain 0.805 for IG and 0.728 for Grad-CAM++. This partial stability may reflect unchanged early visual features and an intact decoder, so it is weaker evidence than the full-randomization result. The check uses only one image and two methods and is consequently diagnostic rather than conclusive.

Attention Rollout and CAV projection are fastest at 0.114 and 0.126 s/image. Grad-CAM++ needs 0.308 s, whereas IG and occlusion take 3.38 and 3.99 s. The complete 12-image experiment remains practical on an A100, but occlusion’s 24.78 GB peak makes it the least portable method. These resource measurements matter when scaling explanations beyond a curated subset.

5 DISCUSSION AND CONCLUSION

Our central finding is that explanation quality is multidimensional. IG and decoder Grad-CAM++ are strongest for identifying regions associated with prediction error, but neither strongly aligns with true depth edges. Attention Rollout is fast and exposes global transformer context, yet its sparse attention should not be interpreted causally. Occlusion supplies the clearest intervention semantics but is coarse and expensive. CAV projection offers a concept-oriented view and the highest deletion effect; however, high variance and the absence of standard TCAV directional-derivative/significance testing limit the strength of this conclusion.

Several design choices constrain the study. First, the global-mean target may suppress object-specific explanations and lets broad image changes dominate deletion scores. Region- or object-conditioned targets would answer more local questions. Second, per-image scale-shift alignment makes the prediction comparison appropriate for relative depth, but it hides absolute-scale errors. Third, only 12 images receive all five explanations, the random baseline is sampled once per fraction, and the randomization test covers one image. Finally, GT-derived concept selection tests representational alignment but does not establish causal concept use.

Dual-subset execution. The pipeline was implemented to run independently on both NYU Depth V2 subsets defined in Section 2: Subset 1 (concept/reference indices 0–794, test

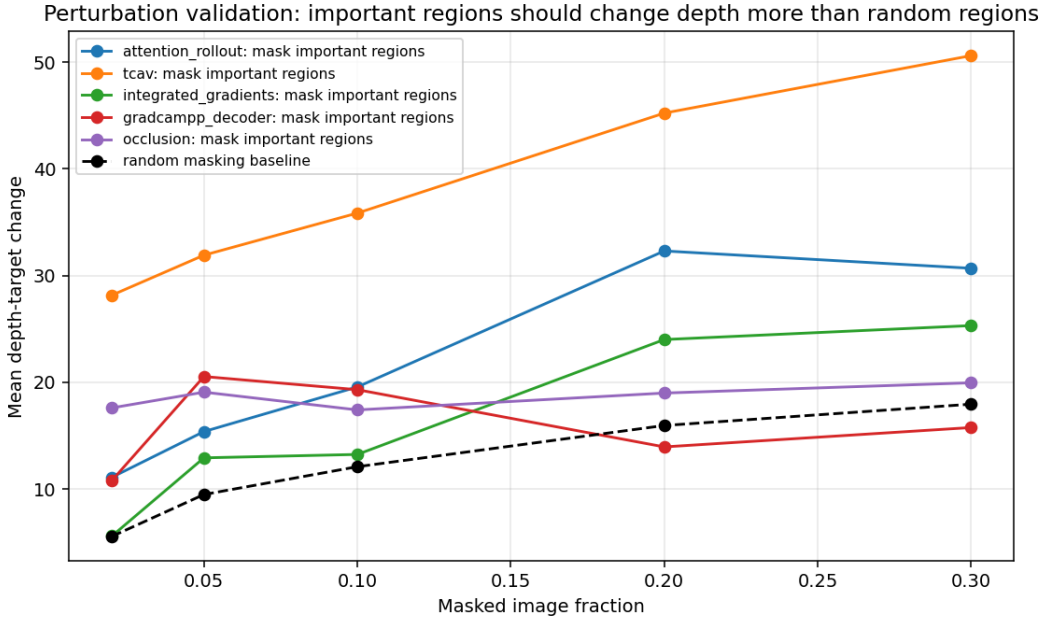


Figure 6: Mean target change as increasingly important regions are replaced by a blurred baseline. CAV projection and Attention Rollout separate most clearly from random masking, though CAV variance across images is high.

indices 795–1448) and Subset 2 (concept/reference indices 1449–2243, test indices 2244–2897). We executed the full notebook sequence twice, once per subset, using separate Jupyter notebooks (`depth_anything_v2_xai_large_model_subset1.ipynb` and `depth_anything_v2_xai_large_model_subset2.ipynb`) so that each run writes its own outputs without overwriting the other. This report presents and discusses **Subset 1 only**. The second run was carried out as a reproducibility check: the aggregate prediction metrics, method rankings, and qualitative XAI patterns were effectively identical across subsets, suggesting that the explanation behavior observed here is stable under the report’s split definition rather than an artifact of one particular index partition. Because the Colab NYU release contains only 1,449 physical image pairs, Subset 2 global indices are mapped to local files by modulo; identical aggregate outcomes therefore reflect consistent pipeline behavior on the same underlying data under different global index labels, not independent new scenes. Full Subset 2 tables and figures are retained in the shared artifact bundle for verification but are omitted here to avoid redundant presentation.

Future work should evaluate more randomly sampled test images with bootstrap confidence intervals, average perturbation results over multiple masks, and compare global, region, and object targets. Standard TCAV scores with repeated random concepts and significance tests should replace the current spatial projection. Cross-dataset tests and comparisons with smaller DAV2 variants would reveal whether explanation behavior transfers or changes with model capacity. Despite these limitations, combining qualitative maps with alignment, perturbation, sanity, and cost measurements provides a substantially more defensible account of model behavior than any single saliency visualization.

Artifact availability. The source notebooks (Subset 1 and Subset 2), generated tables, and report assets are intended to be shared at https://drive.google.com/REPLACE_WITH_SHARED_FOLDER_LINK. **Replace the Google Drive URL before submission.** The DAV2-L checkpoint is available from the official Hugging Face repository, and NYU Depth V2 is downloaded from the official dataset host; full links and reproduction steps are listed in the accompanying README.

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